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A Deep Learning Model for Real-time Detection and Prediction of Moisture Content in the White Tea Withering Process Based on Near-Infrared Spectroscopy

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Abstract: This study proposes a novel deep learning model, named STA-BiGRU-XGBoost, for predicting moisture content during the white tea withering process using near-infrared spectroscopy. The model integrates spatiotemporal attention mechanisms, bidirectional gated recurrent units (BiGRU), and the XGBoost algorithm to address challenges such as extended withering durations, environmental variability, and temporal variations in spectral data. The Maximum Relevance Minimum Redundancy (mRMR) algorithm is employed to select critical variables, including hot air velocity, air duct temperature, and spectral absorbance. A spatial attention mechanism enhances feature relevance, while BiGRU captures long-term temporal dependencies. Temporal attention further adjusts the weights of key time steps. XGBoost is incorporated to improve model robustness against noise. Experimental results using a production-line dataset demonstrate that the STA-BiGRU-XGBoost model achieves superior prediction performance, with RMSE = 0.0920, MAE = 0.0772, and R^2 = 0.9806, significantly outperforming traditional models. Furthermore, the model's interpretability is validated through attention weight visualization, highlighting key features associated with moisture evaporation dynamics.

Keywords: white tea withering; real-time moisture content detection and prediction; near-infrared spectroscopy; deep learning prediction model; spatiotemporal attention mechanism

1. Introduction

White tea, one of China's six major tea types, is renowned for its minimal processing and natural flavor. Its quality is largely determined during the withering process, in which tea leaves undergo moisture loss alongside complex biochemical transformations. These changes significantly influence flavor, aroma, and overall quality, making moisture content a crucial indicator of withering degree and product stability. Moreover, moisture content affects the

conversion of polyphenolic compounds and the retention of amino acids, which contribute to the characteristic flavor and aroma of white tea [1-3].

The withering process typically lasts 48–72 hours and is influenced by factors such as raw material variety, harvest season, leaf maturity, and initial moisture content. Fluctuations in moisture levels can lead to substantial variations in water loss rates and enzymatic activity, ultimately affecting product consistency. Currently, the assessment of withering degree relies primarily on subjective observations, such as leaf softness, color changes, and aroma, which are insufficient for large-scale standardized production [4]. Therefore, a real-time, non-destructive, and accurate method for monitoring moisture content during withering is needed.

Traditional moisture content measurement methods, such as oven drying and Karl Fischer titration, offer high accuracy but are time-consuming, destructive, and unsuitable for online monitoring. Recent advancements in non-destructive testing, particularly near-infrared spectroscopy (NIRS), have demonstrated considerable potential for real-time moisture detection in agricultural products due to their sensitivity to molecular vibrations, including those of water molecules [5]. NIRS has been successfully applied in processes such as green tea fixation, black tea withering, and tea drying, although challenges remain for prolonged processes like white tea withering [6,7].

However, unlike relatively short and concentrated processing stages such as green tea fixation or drying, white tea withering is characterized by a long duration, significant dynamic changes in environmental conditions, and strong process non-stationarity. Continuous variations in environmental temperature, humidity, ventilation conditions, and leaf pile status during withering lead to considerable temporal variations in both moisture migration within tea leaves and their near-infrared spectral responses. Recent studies indicate that during extended processing periods, moisture migration and environmental fluctuations can cause nonlinear drift and dynamic instability in near-infrared spectra, significantly undermining the predictive stability of models based on static assumptions [8,9]. Particularly in long-duration monitoring scenarios such as tea withering, spectra collected at different times may exhibit substantial differences even at similar moisture levels, imposing higher demands on model generalization and robustness [10].

To address these challenges, existing studies primarily employ traditional chemometric methods such as Partial Least Squares Regression (PLSR) and Support Vector Regression (SVR), or spectral correction techniques like External Parameter Orthogonalization (EPO) and Generalized Least Squares Weighting (GLSW) to mitigate environmental disturbances [11,12]. However, these methods generally rely on stable operational assumptions or additional environmental parameter inputs, making them difficult to implement consistently over the long term in actual white tea withering production lines. Furthermore, traditional modeling approaches struggle to simultaneously capture the nonlinear characteristics of moisture

information and its temporal evolution in high-dimensional spectral data, limiting their applicability to prolonged withering processes.

In recent years, deep learning methods have gained significant attention in spectral analysis and food processing monitoring. Convolutional Neural Networks (CNNs) can automatically extract multi-scale spectral features, while Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM), have demonstrated strong capabilities in capturing temporal dependencies in time-series data [13,14]. To further enhance model adaptability under complex conditions, researchers have incorporated attention mechanisms to dynamically assign importance to different time steps or feature dimensions, effectively suppressing redundant information caused by environmental disturbances and improving prediction accuracy and robustness [15-17]. Previous studies have validated the potential of deep learning methods for moisture detection in tea, black tea withering, and agricultural product drying under dynamic conditions [1,3,18]. Recent reviews indicate that combining near-infrared spectroscopy with deep learning represents a promising direction for online quality monitoring in food processing, especially in non-stationary and disturbance-prone industrial applications [19,20]. Nevertheless, for the typical long-duration and environmentally coupled white tea withering process, spectral prediction studies integrating temporal modeling with attention mechanisms remain limited, underscoring the need for further systematic investigation.

Based on the aforementioned challenges and recent research progress, this study focuses on the long-duration, environmentally coupled white tea withering process and aims to develop a near-infrared spectral deep learning prediction framework that integrates temporal modeling with attention mechanisms. The main contributions of this paper are as follows:

(1) A systematic analysis of the correlation between dynamic moisture changes during white tea withering and near-infrared spectral features.
(2) The novel proposal of a STA-BiGRU-XGBoost model that integrates spatiotemporal dual attention mechanisms, bidirectional GRU, and XGBoost to simultaneously enhance long time-series modeling, focus on critical information, and improve robustness against industrial noise.
(3) A comprehensive validation of the proposed method's prediction accuracy, stability, and interpretability in an actual production environment, thereby providing a reliable theoretical foundation and feasible technical support for intelligent and precise control of the white tea withering process.

2. Materials and Methods

2.1 Data Acquisition Device

The white tea withering system comprises several key components, including a feed inlet,

greening mechanism, conveyor belt, dehumidifying heat pump, exhaust fan, discharge outlet, and an infrared spectrometer, as illustrated in Figure 1. Fresh tea leaves are introduced through the bottom inlet and transported by the greening mechanism onto the withering conveyor belt. The dehumidifying heat pump supplies temperature- and humidity-controlled hot air to the withering chamber, facilitating moisture removal while suppressing green odor formation. After 24–48 hours of withering, the leaves are transferred to the discharge outlet, completing the process. The infrared spectroscopy moisture detection system, utilizing a MERS Measurement MS-580 near-infrared moisture analyzer, collects real-time, non-destructive moisture data from the tea leaves. The system operates within the 940–1550 nm wavelength range and is installed directly above the conveyor belt.

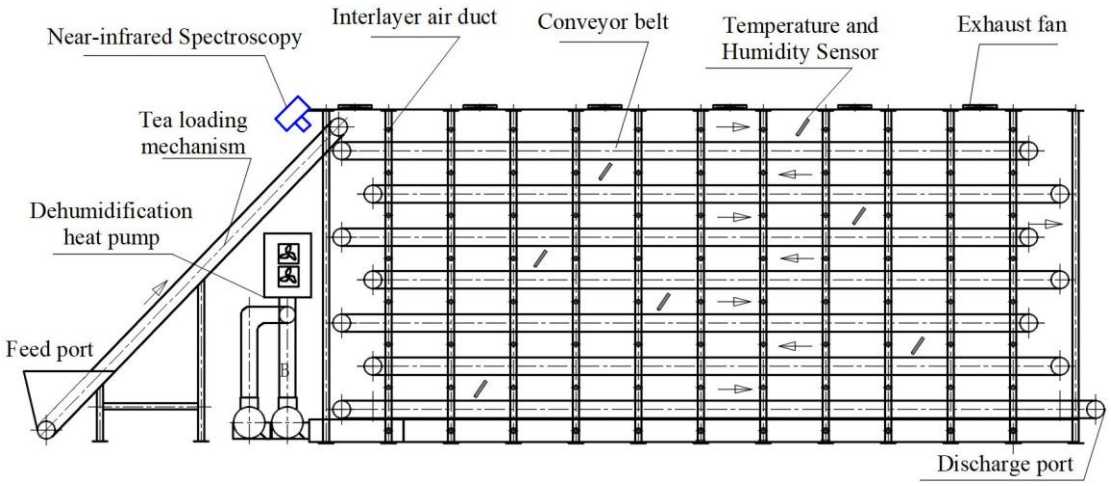
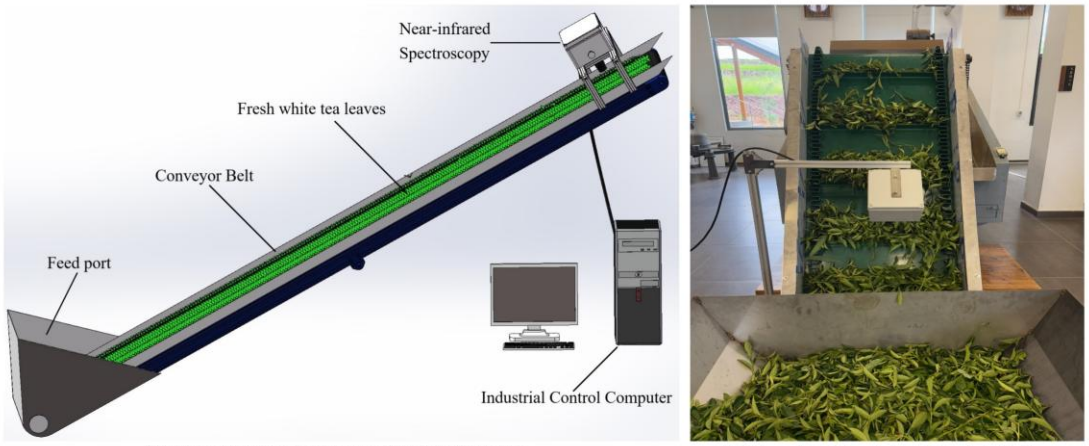


Figure 1. Schematic of the White Tea Withering System

2.2 Sample Preparation and Data Collection

Data collection was conducted from March 18 to May 18, 2025, at a white tea processing company in Wuyishan City. The fresh tea leaves used in the experiments were sourced from Zhenghe County, Nanping City, China. Fresh white tea leaves are primarily harvested between March and May each year, and most tea factories concentrate their withering processes during this period. It should be noted that tea leaves are not withered continuously throughout the year, resulting in minimal seasonal variations in the withering datasets. During tea production, the installed infrared spectrometer collected sample data from leaves in the greening mechanism. The near-infrared spectral data collection method is illustrated in Figure 2. To assess the influence of operational temperature and humidity at different times on near-infrared spectra and prediction accuracy, a sampling interval of 5 minutes was adopted. To ensure consistent withering quality, the tea factory regulated the production workshop environment (ambient temperature: 20–35°C, humidity: 45–60% RH), maintaining stable conditions without significant fluctuations. Withering process data were collected by an industrial control computer (with a sampling frequency of 10 Hz for each sensor, processed using a 5-minute

moving average filter). The computer saved the processed data at the same interval, enabling real-time monitoring and precise control of variables throughout the withering process and ensuring that the collected data accurately reflected actual conditions. Table 1 provides detailed descriptions of the collected process variables, each carefully selected based on its significant impact on final quality and processing efficiency. The dataset was divided into three subsets: a training set for learning model parameters, a validation set for hyperparameter tuning, and a test set comprising two batches for evaluating prediction model performance.



(a) Schematic of Near-Infrared Spectral Data Collection

(b) Schematic of Near-Infrared Spectral Working Environment Data Collection

Figure 2. Schematic of Near-Infrared Spectral Data Collection

Table 1. Withering Tea Process Variables.

Serial No.	Variable names	Unit	Scope of the variables
X_1	Withering outdoor temperature	℃	10.0-40.0
X_2	Withering outdoor humidity	%	50.0-90.0
X_3	Withering indoor temperature	℃	20.0-40.0
X_4	Withering indoor humidity	%	50.0-80.0
X_5	Withering room heat pump outlet temperature	℃	20.0-50.0
X_6	Withering room hot air velocity	m/s	0-3.0
X_7	Withering room interlayer air duct outlet temperature	℃	20.0-40.0
X_8	Feed flow rate	kg/s	0-10.0
X_9	Near-infrared spectral absorbance	%	10-150
X_{10}	Discharge port temperature	℃	20.0-40.0

2.3 Measurement of Moisture Content

The moisture content of fresh white tea leaves was determined using the direct drying method as specified by the Chinese National Standard GB5009.3-2016 [21], the measurement of moisture content is shown in Figure 3.. Each sample was weighed before and after drying in an oven. The average moisture content from three measurements was used to calculate the moisture percentage with the following formula:

$$w=\frac{m-m_1}{m}\times 100\% \quad (1)$$

where m is the initial mass of the tea sample and m_1 is the mass after drying.



Figure 3. The Measurement of Moisture Content

2.4 Data Preprocessing

Due to external environmental factors and instrument interference, near-infrared spectroscopy systems are susceptible to noise. Therefore, appropriate spectral preprocessing techniques are crucial for mitigating these effects. In this study, four spectral preprocessing algorithms were applied: Multiplicative Scatter Correction (MSC), Standard Normal Variate (SNV), Savitzky–Golay smoothing (SG), and first-order derivative (D1). SNV and MSC correct for optical path variations, reducing scattering effects on spectral information. SG and D1 remove noise and reduce spectral errors caused by derivative-based background effects. These preprocessing methods help diminish external interference, improving prediction model accuracy and generalization. Figure 4 shows the raw near-infrared spectral curve of a fixed leaf sample. Significant absorption peaks were observed in the 1160–1220 nm and 1420–1480 nm bands, primarily associated with symmetric and asymmetric stretching vibrations of O–H bonds in water molecules [22,23]. After MSC preprocessing, absorption peaks in training samples with different moisture gradients became clearer with reduced scattering, enabling more accurate

moisture content prediction, as shown in Figure 5.

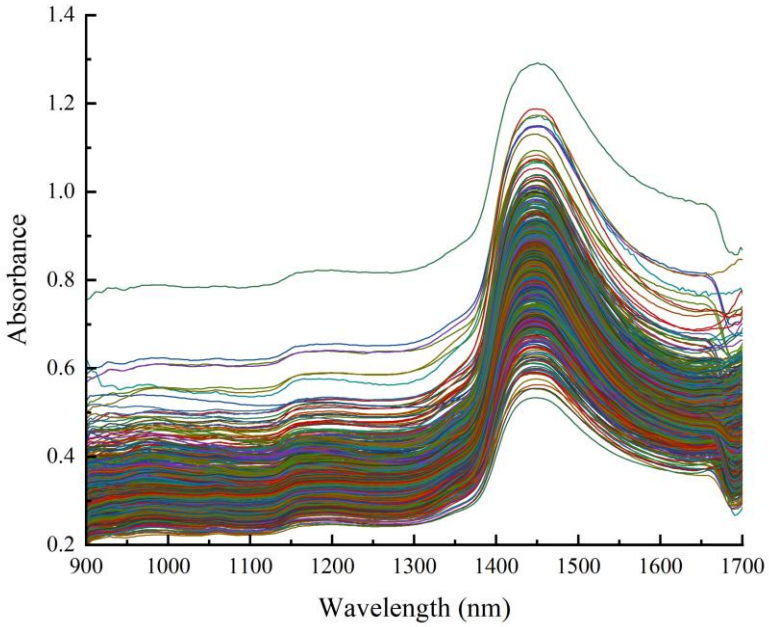


Figure 4. Raw NIR spectral curves of white tea leaves

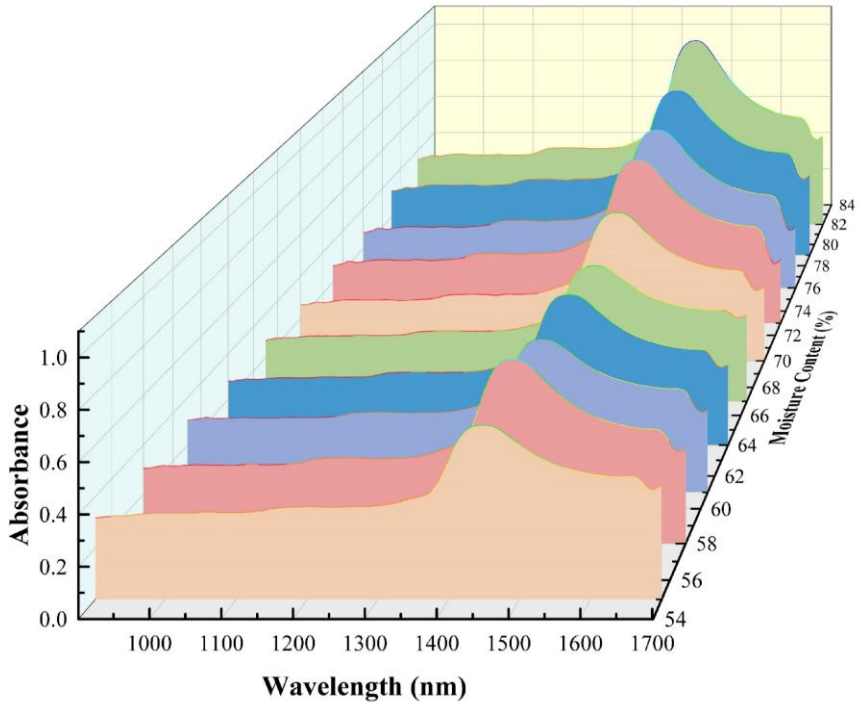


Figure 5. NIR spectral curves after MSC preprocessing

2.5 Prediction Method Determination

The white tea withering process occurs in a highly variable field environment where sensor sensitivity can be affected by external disturbances. To address this, we employed a neural

network to process infrared spectrometer data. Neural networks can be categorized into feedforward and recurrent types. Feedforward networks, lacking feedback and inhibition mechanisms, are unsuitable for this task. In contrast, recurrent networks, which incorporate feedback and inhibition, optimize the network through algorithms, making them more appropriate for this study.

Deep learning networks, especially those designed for time-series prediction, have demonstrated outstanding performance in modeling complex dynamic processes. In this study, the STA-BiGRU model was applied for the first time to predict moisture content during tea withering. The model integrates STA and BiGRU, enabling adaptive weighting of time steps and capturing long-term dependencies in moisture content sequences.

The withering process exhibits dynamic characteristics, including spatial and temporal dependencies, along with noisy and inconsistent data, which pose challenges for accurate prediction. The data characteristics can be summarized as follows:

- **Spatial dependence:** Heat and mass transfer phenomena lead to significant spatial correlations among variables.
- **Temporal dependence:** Moisture content is influenced by past values, and fluctuations in process parameters cause dynamic changes.
- **Data quality:** The complex environment results in noisy, missing, and inconsistent data, affecting model performance.

Given a set of input variables $X = \{x_1, x_2, \dots, x_t\}$, where t represents time and x_t is the observed value at time t, the model predicts the key variables for future time periods. The prediction function can be expressed as:

$$[\hat{y}_{T+1}, \hat{y}_{T+2}, \cdots, \hat{y}_{T+\tau}] = \hat{f}(X, y_1, y_2, \cdots, y_T) \quad (2)$$

where $\hat{f}$ is a nonlinear prediction function, T represents the size of the historical data window, τ is the forecast horizon, y_T is the moisture content at time T, and $\hat{y}_{T+1}$ is the predicted moisture content at time $T + 1$.

2.6 STA-BiGRU Method

The STA-BiGRU method for moisture content prediction during white tea withering consists of four main components: feature selection using mRMR, a spatiotemporal attention mechanism, bidirectional GRU modeling, and XGBoost-based moisture content prediction (MC). The overall framework is illustrated in Figure 6. The key steps are as follows:

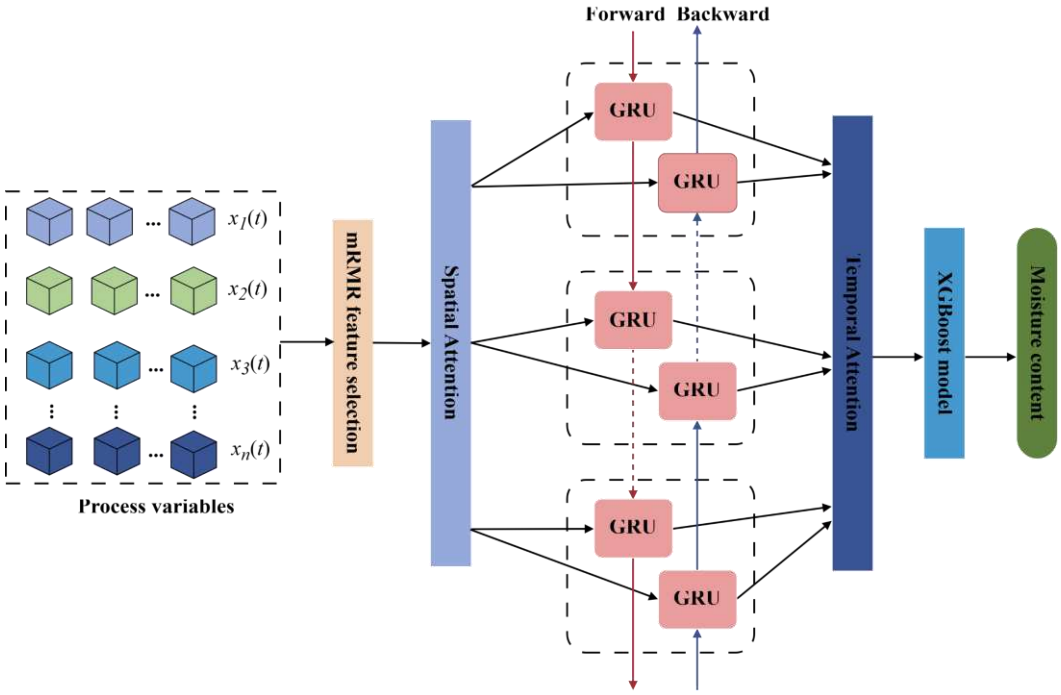


Figure 6. Overall Framework of the STA-BiGRU-XGBoost Prediction Model

During tea withering, high-dimensional variables collected by the processing system (e.g., environmental temperature, humidity, spreading thickness, and ventilation rate) increase model complexity and may reduce prediction accuracy. Previous studies have shown that the mRMR algorithm effectively removes redundant features while retaining key variables, improving model performance (with a 9% increase in R^3) and preserving the physical interpretability of selected features[24,25]. Therefore, the mRMR algorithm was adopted for feature selection in this study.

During the tea withering process, different variables exert varying levels of influence on moisture content at different time points. A spatial attention mechanism was employed to adaptively assign weights to each feature according to its correlation with moisture content, thereby enhancing the model's ability to focus on relevant variables. Let the input variables be denoted as $X = \{x_1(t), x_2(t), \ldots, x_n(t)\}$, where $t$represents time, $x_n(t)$denotes the observed value at time t, and n is the number of input features. For each time step t, the attention weight $a_i$of each feature is calculated as follows:

$$a_i = \text{softmax}(e_i) = \frac{\exp(e_i)}{\sum_{i=1}^{m} e_i} \tag{3}$$

$$e_i = \text{sigmoid}(W_s x_n + b_s)$$

where x_n is the input vector of the n-th feature at time step t, W_s, and b_s are the parameters of the spatial attention layer, which compute the weighted sum of the input features as the context vector X':

$$X^{'}=\mathrm{A}\odot X \tag{4}$$

where A is the attention weight vector and $\odot$ denotes the Hadamard product. The spatially weighted input features X' serve as the input to the bidirectional GRU at time step t, enabling the model to focus on features most relevant to the current moisture content prediction.

GRU captures long-term dependencies in time-series data through update and reset gates, making it suitable for handling temporal correlations in tea withering. This study employs BiGRU, which combines forward and backward information, enhancing the model's understanding of sequential data. The key equations of GRU are as follows:

$$\begin{cases} z_t=\sigma(W_z[h_{t-1},X_t']+b_z) \\ r_t=\sigma(W_r[h_{t-1},X_t']+b_r) \\ \tilde{h}_t=\tanh(W_h[\mathrm{r}_t\odot h_{t-1},X_t']+b_h) \\ h_t=(1-z_t)\odot h_{t-1}+z_t\odot\tilde{h}_t \end{cases} \tag{5}$$

where z_t is the update gate controlling historical state retention, r_t is the reset gate adjusting historical information forgetting, h_{t-1} and h_t are previous and current hidden states, X_t' is the spatially weighted input, $\sigma(\cdot)$ is the sigmoid activation function, and $\odot$ is the Hadamard product.

Standard GRUs only consider forward information flow. This study uses BiGRU to capture long-term temporal dependencies in withering. The bidirectional structure allows the model to utilize both historical and future information, improving its ability to capture complex temporal dependencies. BiGRU consists of two GRU sub-networks running in opposite directions, capturing forward and backward dependencies, respectively. By concatenating forward and backward hidden states, BiGRU comprehensively represents moisture variation dynamics during withering:

$$\begin{cases} \vec{\mathbf{h}}_t=\overrightarrow{\mathrm{GRU}}(\mathbf{h}_{t-1},X_t') \\ \overleftarrow{\mathbf{h}}_t=\overleftarrow{\mathrm{GRU}}(\mathbf{h}_{t+1},X_t') \\ \mathbf{H}_t=[\vec{\mathbf{h}},\overleftarrow{\mathbf{h}}] \end{cases} \tag{6}$$

where $\vec{h}_t$ and $\overleftarrow{h}_t$ are outputs of the forward and backward GRUs, respectively, and X_t' is the initial cell state (zero-initialized).

The tea withering process exhibits significant long-term dependencies and dynamic characteristics. As input sequence length increases, model performance tends to degrade. To address this, a spatiotemporal attention mechanism is employed to adaptively weigh the importance of different time steps. Let $H=[h_1,h_2,\ldots,h_n]$ represent hidden states from BiGRU; the temporal attention weight β_i for each time step is calculated as:

$$l_i=ReLU(W_i\mathrm{h}_i+b_i) \tag{7}$$

$$\beta_i=soft\max(l_i)=\frac{e^{l_i}}{\sum_{i=1}^{n}l_i} \tag{8}$$

where W_i is a trainable weight matrix, b_i is the bias, and ReLU is the activation function. The context vector H' is computed as the weighted sum:

$$H^{'}=B\odot H \tag{9}$$

where $B = [\beta_1, \beta_2, \dots, \beta_n]^T$. The context vector H' summarizes the most relevant temporal information for final prediction.

2.7 mRMR Feature Selection for Tea Withering Process Variables

The tea withering system is influenced by numerous process variables; however, high-dimensional data can complicate network structure and reduce prediction accuracy. To address this, the mRMR algorithm was used to extract informative and highly correlated features from high-dimensional inputs. Given an input feature set $S = \{x_1(t), x_2(t), \dots, x_n(t)\}$ and target variable $y(t)$, mRMR selects a feature subset that maximizes relevance to the target while minimizing redundancy. The objective function is:

$$mRMR = \max_S \left[\frac{1}{|S|} \sum_{X_i(t) \in S} I\big(X_i(t) : Y(t)\big) - \frac{1}{|S|^2} \sum_{X_i(t), X_j(t) \in S} I\left(X_i(t) : X_j(t)\right) \right] \tag{10}$$

where $I(\cdot)$ denotes mutual information. Features are incrementally added to maximize the objective until K features are selected, with K tuned via validation performance.

Applying mRMR to variables in Table 1, X_6 (hot air velocity), X_7 (air duct outlet temperature), and X_9 (spectral absorbance) showed the highest correlations and were selected as inputs for the deep learning model.

2.8 Spatial Attention Method

Strong coupling exists among process variables during withering, and their importance to moisture content changes dynamically. A spatial attention method adaptively assigns weights to variables at each time step based on their correlation with moisture content. By dynamically focusing on relevant features, the model better captures spatial dependencies and interactions. The spatial attention mechanism is illustrated in Figure 7.

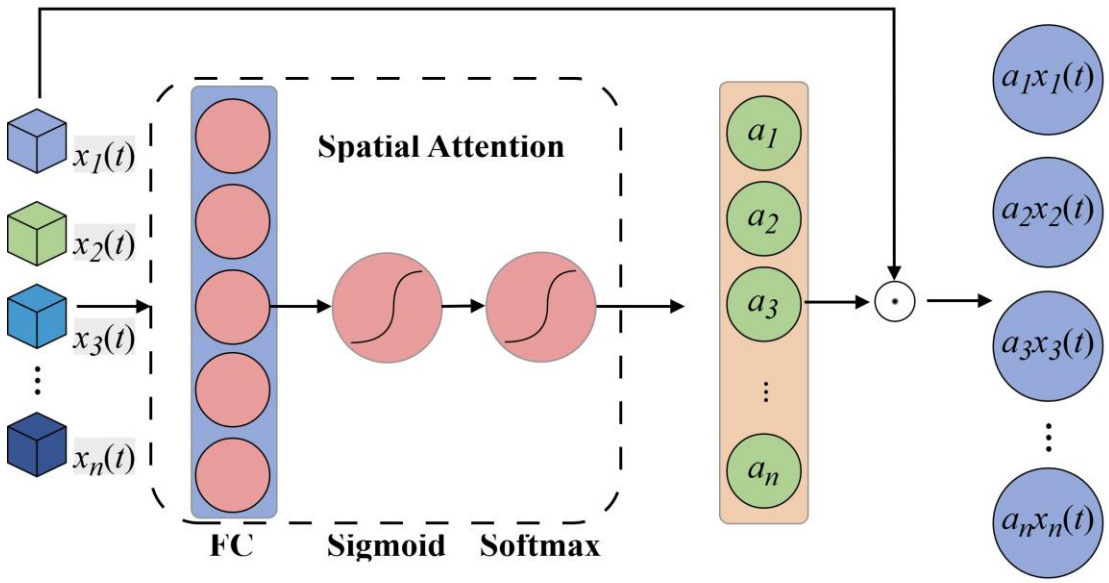


Figure 7. Schematic diagram of the spatial attention method

For input variables $X = \{x_1(t), x_2(t), \ldots, x_n(t)\}$, the attention weight a_i for each feature at time t is calculated as:

$$a_i = softmax(e_i) = \frac{exp(e_i)}{\sum_{i=1}^{m} e_i} \tag{11}$$

$$e_i = sigmoid(W_s x_n + b_s) \tag{12}$$

where x_n is the input vector of the n-th feature, and W_s, b_s are spatial attention parameters. The context vector X' is:

$$X' = A \odot X \tag{13}$$

where $A = [a_1, a_2, \ldots, a_n]$. Spatially weighted features X' are fed into BiGRU at time step t.

2.9 GRU Prediction Model

GRU is a recurrent network composed of reset and update gates, efficiently controlling information storage and updating in time-series data. Its structure is shown in Figure 8.

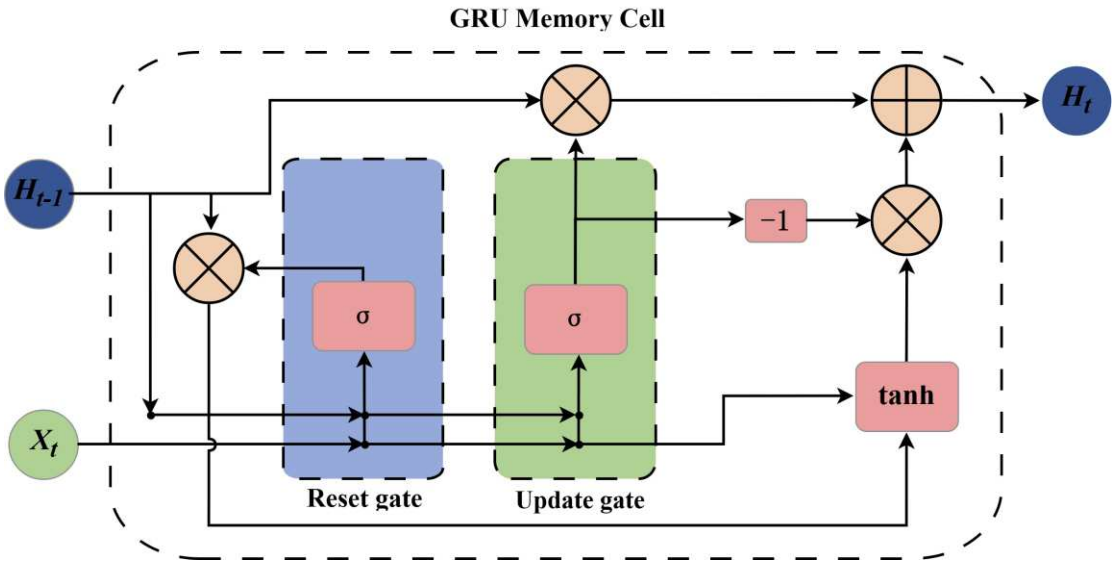


Figure 8. Schematic diagram of the GRU.

Traditional GRUs process information in one direction. This study uses BiGRU to capture long-term temporal dependencies by utilizing both past and future contexts. The BiGRU structure (Figure 9) comprises two GRU layers running opposite directions.

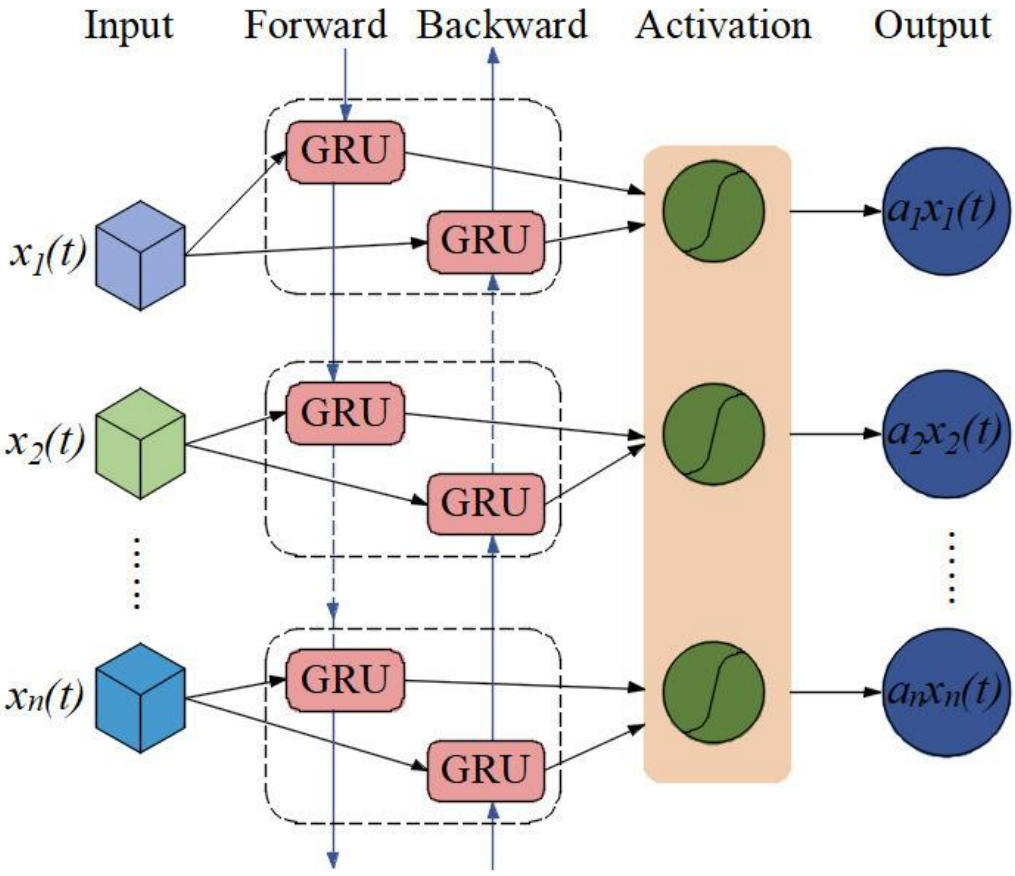


Figure *9*. The structure of BiGRU

The final BiGRU output is the concatenation of forward and backward hidden states, as in Equation (6).

2.10 Time Attention Method

To handle long-term dependencies and input sequence length issues, a temporal attention mechanism adaptively weights different time steps based on their relevance to current prediction. The mechanism is illustrated in Figure 10.

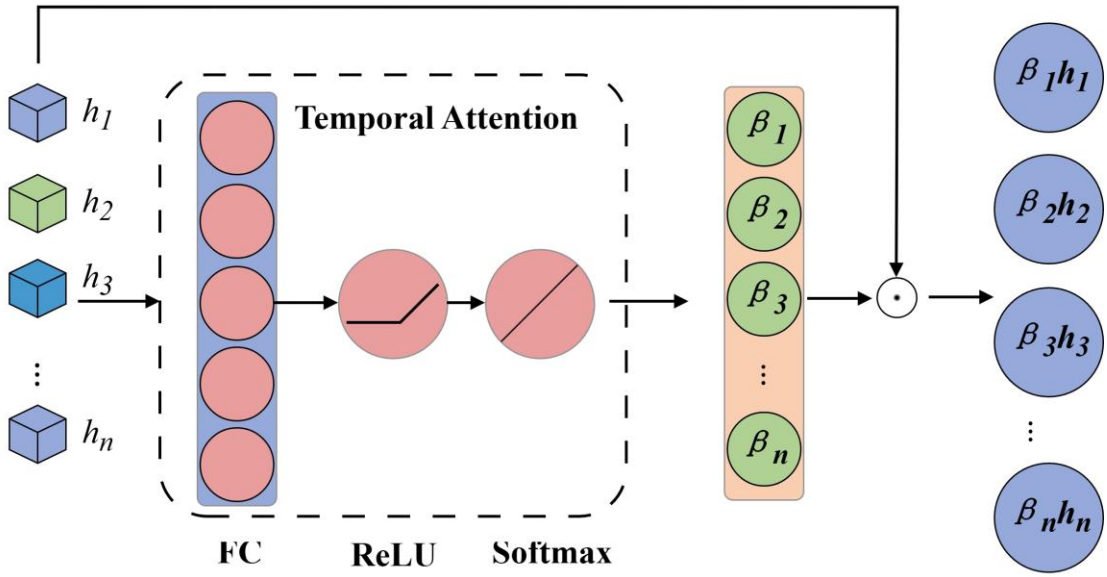


Figure *10*. The schematic diagram of the temporal attention method

For hidden states $H = [h_1, h_2, \dots, h_n]$ from BiGRU, temporal attention weight β_i is:

$$l_i = ReLU(W_i h_i + b_i) \tag{14}$$

$$\beta_i = soft\,max(l_i) = \frac{e^{l_i}}{\sum_{i=1}^{n} l_i} \tag{15}$$

where W_i is the training weight matrix, b_i is the bias term, and ReLU is the activation function.

The context vector H' is:

$$H' = B \odot H \tag{16}$$

where $B = [\beta_1, \beta_2, \dots, \beta_n]^T$. This vector summarizes relevant temporal information for final prediction.

2.11 XGBoost Algorithm

To enhance generalization and robustness against noise and harsh conditions, XGBoost is integrated as a post-processing step after STA-BiGRU. XGBoost is an efficient ensemble

learning algorithm [26] that accelerates training via multithreading and distributed computing. It optimizes model parameters by minimizing a loss function comprising training error and regularization:

$$Obj(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{17}$$

where ϕ denotes model parameters, l is the loss function, and $\Omega(f_k)$ regularizes the k-th tree:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} \omega_j^2 \tag{18}$$

with T leaf nodes, leaf weights ω_j, leaf penalty γ, and L2 regularization λ. Minimizing this objective reduces overfitting and improves generalization.

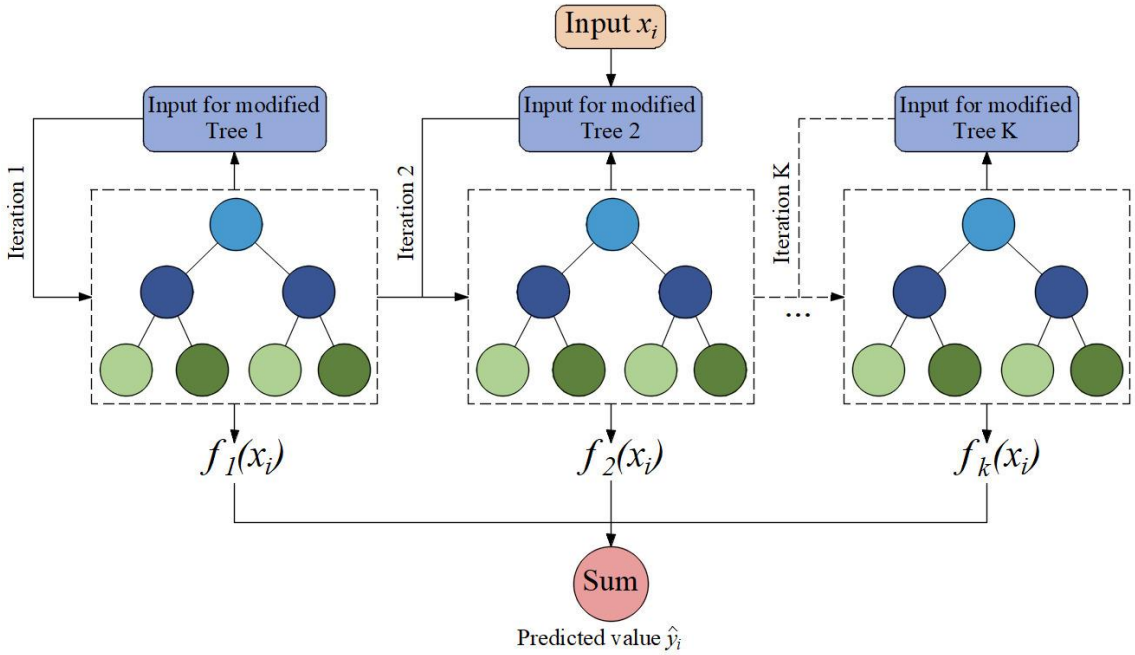


Figure 11. Diagram of the XGBoost algorithm

XGBoost is applied to the output of the attention-enhanced BiGRU (Figure 11) to handle noise and outliers common in withering environments. The final prediction is:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\boldsymbol{H}') \tag{19}$$

where H' represents the context vector from the attention mechanism, and K represents the number of trees.

2.12 Model Evaluation Metrics

Three metrics evaluate model performance: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2):

$$RMSE = \sqrt{\left(\sum_{j=1}^{n} \left(y_j - \hat{y}_j\right)^2 / n\right)} \tag{20}$$

$$MAE = \sum_{j=1}^{n} |\hat{y}_j - y_j| / n \tag{21}$$

$$R^2 = 1 - \frac{\sum_{j=1}^{n} \left(\hat{y}_j - y_j\right)^2}{\sum_{j=1}^{n} \left(y_j - \bar{y}_j\right)^2} \tag{22}$$

where n is sample count, y_j and $\hat{y}_j$ are true and predicted values, and $\bar{y}$ is the true mean.

3. Results and Discussion

3.1 Experimental Setup

The proposed STA-BiGRU-XGBoost model was compared with several deep learning models:

- **LSTM:** Advanced RNN handling long-term dependencies.
- **GRU:** Simplified LSTM variant with fewer parameters.
- **BiGRU:** Bidirectional GRU capturing past and future context.
- **TA-BiGRU:** BiGRU with temporal attention.
- **STA-BiGRU:** BiGRU with spatiotemporal attention.
- **STA-BiGRU-XGBoost:** STA-BiGRU integrated with XGBoost for noise robustness.

Hyperparameters were tuned via validation performance; configurations are listed in Table 2.

Table 2. Hyperparameter Configurations for All Comparative Models.

Part A: Deep Learning Backbone Models

Prediction methods	Hidden Units	Dropout	Lr	Batch size	Epochs	Training Strategy
LSTM	256	0.35	0.001	128	100	Early Stopping (Patience=15)
GRU	256	0.35	0.001	128	100	Early Stopping (Patience=15)
BiGRU	256	0.35	0.001	128	120	Early Stopping (Patience=20)

TA-BiGRU	256	0.35	0.001	128	120	Early Stopping (Patience=20)
STA-BiGRU	256	0.35	0.001	128	120	Early Stopping (Patience=25)
STA-BiGRU-XGBoost	256	0.35	0.001	128	120	Early Stopping (Patience=25)

Part B: XGBoost Hyperparameter Tuning (for STA-BiGRU-XGBoost Ensemble)

Hyperparameter	Baseline	Search Space	Purpose
n_estimators	300	[100,200,300,400,500]	Number of boosting rounds
max_depth	6	[3,6,9]	Tree complexity and interaction depth
learning_rate (η)	0.05	[0.05,0.1,0.2]	Shrinks tree contributions
subsample	0.8	[0.7,0.8,0.9,1.0]	Row sampling per tree
colsample_bytree	0.8	[0.7,0.8,0.9,1.0]	Feature sampling per tree
reg_alpha (L1)	0	[0,1e-5,1e-2]	L1 regularization on leaf weights
reg_lambda (L2)	1	[1,1.5,2]	L2 regularization on leaf weights

Notes:

Early Stopping: Applied to all deep learning models based on validation loss plateauing. Patience (epochs) is set per model complexity.

Optimizer: Adam optimizer is recommended for all deep learning models.

XGBoost Tuning: The ensemble model's XGBoost component undergoes independent hyperparameter optimization using the defined search space (e.g., via RandomizedSearchCV) to minimize validation RMSE.

The experiments were conducted on an industrial computer equipped with an Intel i7-1165G7 CPU (2.80 GHz) and 32.0 GB of RAM. The improved STA-BiLSTM method and other algorithms were implemented using TensorFlow 2.19 and Python 3.13.0. The XGBoost model was built using the XGBoost 3.0 library.

3.2 Experimental Results

3.2.1 Prediction Performance Comparison

Table 3 presents experimental results on the real-world withering dataset. STA-BiGRU-XGBoost achieved the best performance with RMSE = 0.0920, MAE = 0.0772, and R^2 = 0.9806, significantly outperforming all baseline models. LSTM and GRU showed higher errors and lower R^2, while BiGRU improved performance by capturing bidirectional context. TA-BiGRU further reduced error via temporal attention. STA-BiGRU, with spatiotemporal attention, achieved lower errors (RMSE = 0.1325, MAE = 0.1099) and higher R^2 (0.9533). The integration of XGBoost enhanced robustness, yielding the lowest errors and highest R^2.

Table 3. Experimental Results on the Tea Withering Dataset

Methods	RMSE	MAE	R^2
LSTM	0.6962	0.5839	0.5162
GRU	0.6558	0.5072	0.6258
BiGRU	0.4824	0.3813	0.6731
TA-BiGRU	0.2465	0.2045	0.8141
STA-BiGRU	0.1325	0.1099	0.9533
STA-BiGRU-XGBoost	0.0920	0.0772	0.9806

3.2.2 Prediction and Error Curve Analysis

Figure 12 compares actual vs. predicted moisture content curves for all models. LSTM and GRU exhibit considerable deviations and poor trend tracking, consistent with large error fluctuations in Figure 13. BiGRU better captures temporal dynamics via bidirectional memory. TA-BiGRU reduces error further by focusing on relevant time steps. STA-BiGRU, with spatial attention, improves performance, reducing prediction error by 11.2% relative to TA-BiGRU. STA-BiGRU-XGBoost shows the closest agreement with actual curves, with an average error reduction of 55.5% compared to STA-BiGRU, meeting on-site assessment requirements.

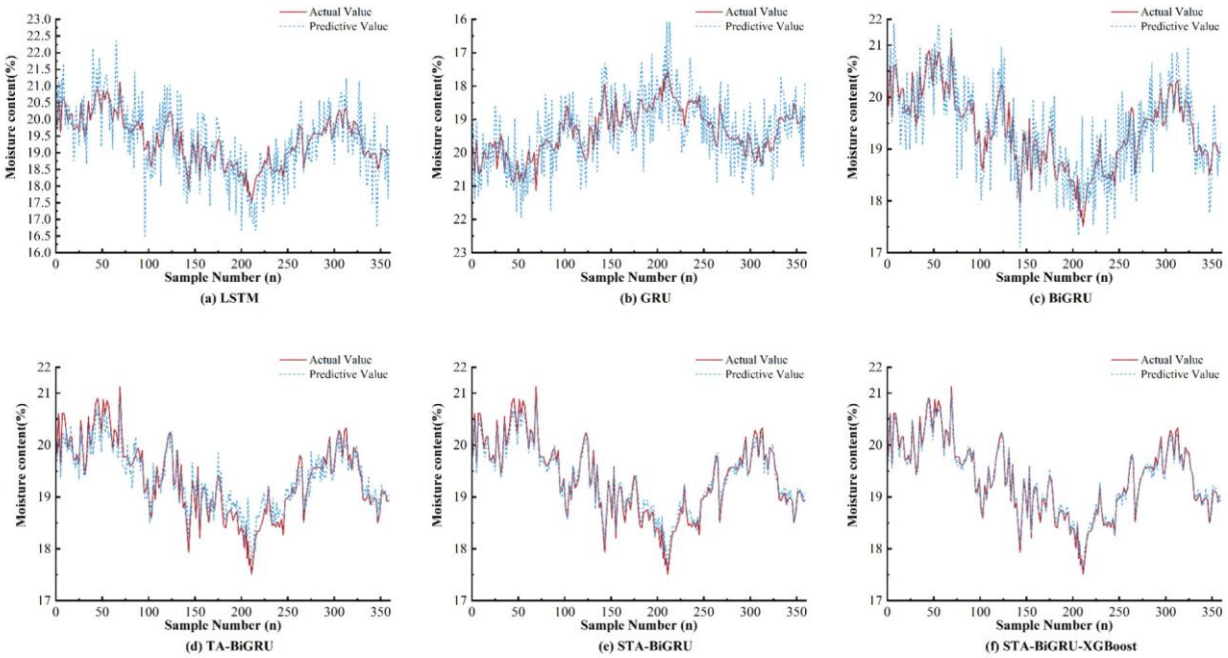


Figure 12. Comparison of different moisture content prediction models

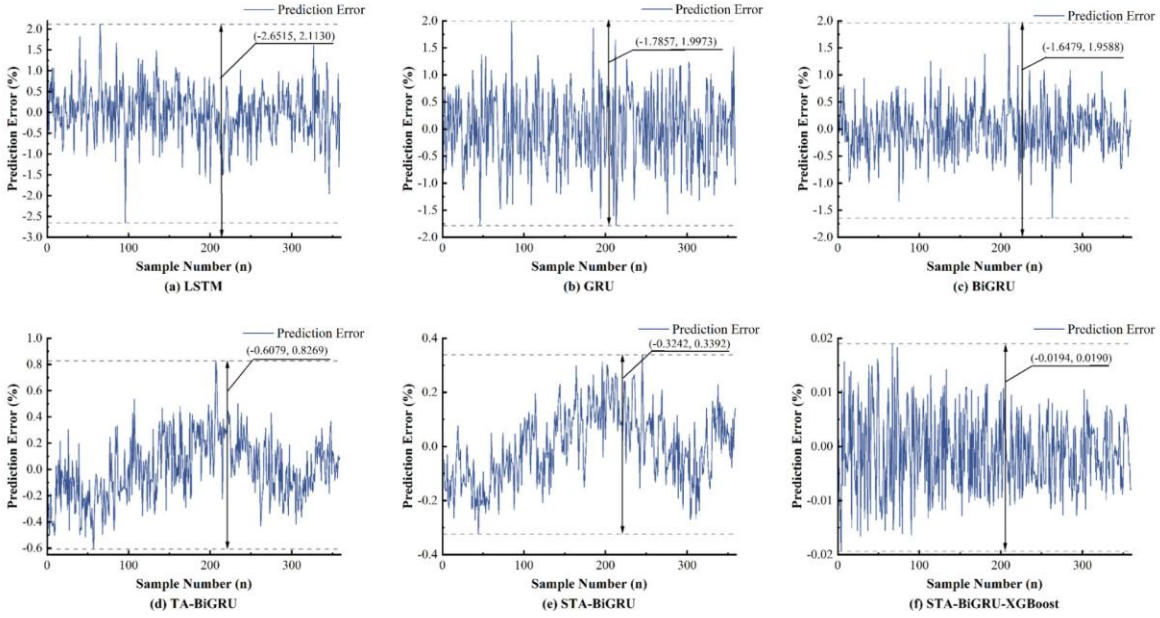


Figure 13. Comparison of the errors of different moisture content prediction models

3.2.3 Robustness and Reliability Assessment

Figure 14 displays prediction error probability distributions for all models. STA-BiGRU-XGBoost exhibits the most concentrated distribution centered near zero, indicating superior reliability. Figure 15 shows error boxplots; STA-BiGRU-XGBoost has the narrowest interquartile range and fewest outliers, confirming exceptional robustness.

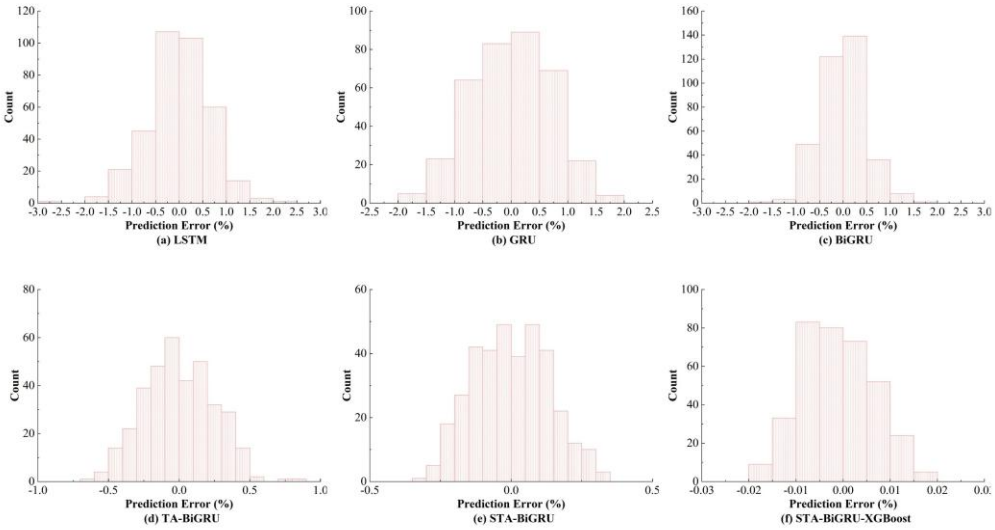


Figure 14. Error probability distribution curves of different moisture content prediction models

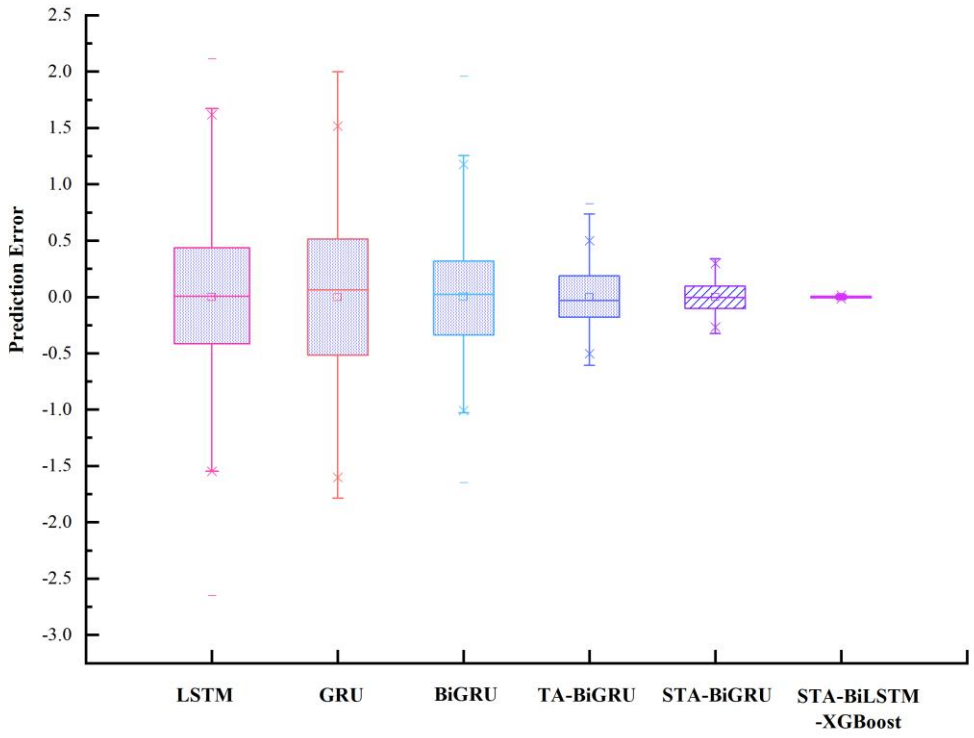


Figure 15. Boxplots of predict error for different models

3.2.4 Validation on a Secondary Test Set

A secondary independent test set validated model reliability (Table 4). Results align with initial findings: attention-based models and the hybrid ensemble outperform conventional ones. STA-BiGRU-XGBoost again achieves the highest accuracy (RMSE = 0.0980, MAE = 0.0643, R^2 = 0.9879), confirming its effectiveness across datasets.

Table 4. Results on Secondary Tea Withering Dataset.

Methods	RMSE	MAE	R^2
LSTM	0.6610	0.5721	0.5058
GRU	0.6414	0.5145	0.6147
BiGRU	0.4712	0.3950	0.6542
TA-BiGRU	0.2314	0.2158	0.8244
STA-BiGRU	0.1348	0.1102	0.9590
STA-BiGRU-XGBoost	0.0980	0.0643	0.9879

3.3 Interpretability of Experiments

To enhance interpretability, attention weight matrices were visualized as heatmaps (Figures 16 and 17). Spatial attention (Figure 16) shows weights across process variables over time. Variables X_6 (hot air velocity), X7 (air duct outlet temperature), and X_9 (spectral absorbance) exhibit consistently high weights (darker hues), indicating critical roles in withering. Variable X8 (feed flow rate) shows lower weights (lighter hues), aligning with known process dynamics. In industrial settings, increased X_6 and X_7 accelerate moisture evaporation, affecting spectral absorption—validating the model's focus on physically meaningful variables.

Temporal attention (Figure 17) illustrates weight distribution across time steps. The current step (t) receives the highest weight (0.46), emphasizing its importance for instantaneous prediction. Weights decay for earlier steps, demonstrating adaptive focus on recent relevant information while retaining past context.

The spatiotemporal attention mechanisms not only improve performance but also provide insight into model decisions. Spatial heatmaps validate alignment with domain knowledge, while temporal distributions reveal how historical data is weighted, enhancing model transparency and credibility.

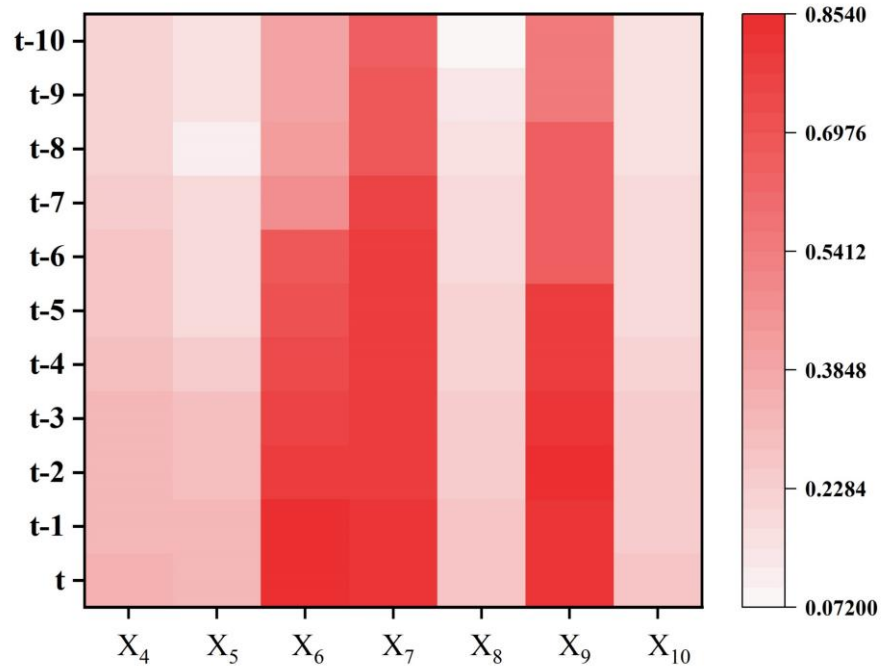


Figure 16. Hot map of the spatial attention weight

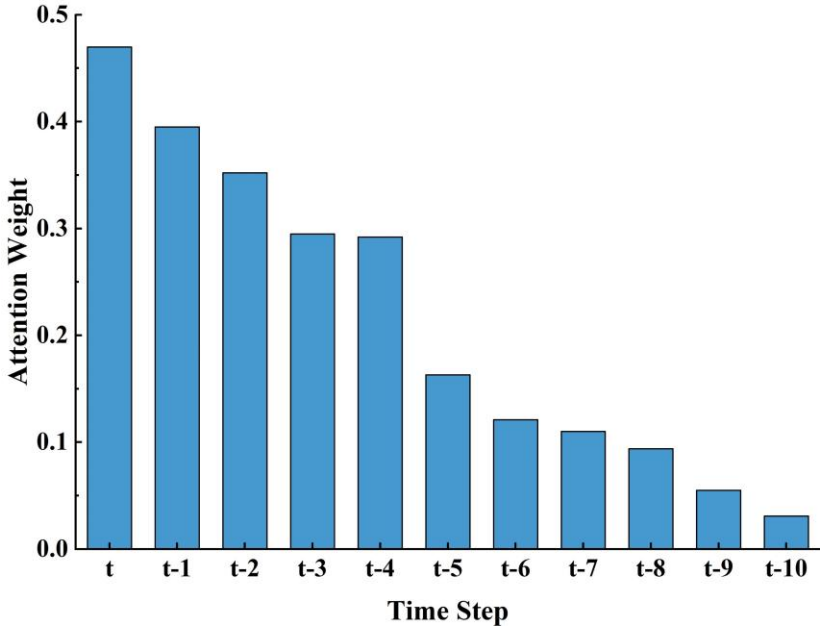


Figure 17. The temporal attention weight

4. Conclusion

This study successfully developed and validated an improved STA-BiGRU-XGBoost deep learning model for high-precision real-time moisture content prediction during white tea withering. The model addresses key challenges in long-duration industrial monitoring via a multi-layered architecture: mRMR feature selection ensures interpretability; spatiotemporal

attention adaptively focuses on relevant variables and time steps; BiGRU captures long-term dynamic dependencies; and XGBoost enhances generalization in noisy environments. Experimental results demonstrate that STA-BiGRU-XGBoost outperforms baseline models (LSTM, GRU, BiGRU, and attention variants) in accuracy (RMSE, MAE), goodness-of-fit (R^2), and error stability. Visualization of spatial attention weights confirms that model decisions align with core physical mechanisms (e.g., hot air velocity and temperature driving moisture evaporation), transforming it from a "black box" into a reliable, transparent, and interpretable tool.

However, the current model is optimized for single-step prediction; its multi-step rolling prediction capability remains unexplored. Future work should: (1) design multi-step prediction architectures and explore decomposition methods for moisture content time series; (2) integrate multimodal data (e.g., visible light, hyperspectral imaging) to comprehensively characterize withering quality transformations; and (3) extend the framework to other agricultural processes with similar long-duration, strongly coupled characteristics to verify generalizability.

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Author Contributions:

Conceptualization, W.T. and B.C.; methodology, W.T. and B.G.; software, X.Y. and L.C.; validation, Y.H., R.L., J.R., and W.T.; investigation, L.C. and J.R.; data curation, Y.H. and B.G.; writing—original draft preparation, W.T. and B.C. All authors have read and agreed to the published version of the manuscript.

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